Spottr:

An AI Member Retention System

Product	AI-powered member churn prediction and retention briefing system
Sector	Boutique fitness: Owner-operated European studios. 80-250 direct members
Stack	Python · scikit-learn · GPT-4o-mini · n8n · Plotly Dash · Tally.so · Fitbit OAuth 2.0
Author	Dina Bosma-Buczynska | AI Consultancy Bootcamp | March 2026

1. Use Case & Problem Definition

1.1 Business Problem

Boutique fitness studios lose approximately 40% of members per year (IHRSA). For a 150-member studio at €120/month average, that is €90,000 in preventable revenue loss annually. Churn is almost never sudden; members disengage over 4-6 weeks before cancelling. The drift is visible in attendance data, but no one is watching closely enough to act in time.

At 30 members, a coach knows everyone. At 150, the members going quiet are precisely the ones who stop being visible. Spottr solves this: automated early warning, delivered to coaches before the window closes.

1.2 Company Profile

Attribute	Detail
Target size	80–250 direct members (not USC/Gympass pass holders - different retention economics)
Revenue	€150,000 - €600,000/year. Owner-operated, 2-8 coaches, no in-house data team.
Technology	Gym management system (Mindbody, Gymdesk, Magicline) Attendance data only, no predictive analytics
GDPR status	Aware of obligations. No DPO, no compliance team. This is a purchase blocker. Spottr removes it.

1.3 The AI-Powered Solution

A scheduled pipeline ingests member visit data, scores every member using a trained logistic regression model, and sends coaches a plain-English briefing email every Monday at 07:00. The AI summarises. The coach decides. The human relationship stays at the centre.

Output	Frequency	Content	Recipients
Weekly email	Monday 07:00	Risk ranking, coaching context per flagged member, survey responses, WATCH alerts	All coaches + owner
Dashboard	Always-on	Full member risk view, colour-coded, sortable, revenue at risk	Studio owner
Monthly summary	1st of month	Satisfaction trends, NPS, low-score alerts, full response table	Studio owner

1.4 Two-Layer AI Architecture

Layer	Component	What it does	Cost / frequency
Layer 1	**scikit-learn logistic regression**	Looks at visit frequency, recency, and tenure. Outputs a churn probability (0-1) per member. No language, no reasoning, just pure statistics.	Free. Runs offline on laptop. Once per week.
Layer 2	**GPT-4o-mini via OpenAI API**	Takes the risk scores for flagged members and writes plain-English coaching notes. Turns numbers into sentences a coach can act on.	~€0.15/run. One batched call. Once per week.

> *Layer 1 decides who to flag. Layer 2 writes what to say about them. Coaches decide what to do.*

GPT-4o-mini runs at temperature=0 (deterministic output). At POC scale, output quality is reviewed manually via the n8n execution log after each run, the full prompt and response are recorded there.

From Phase 2 onwards, LangSmith (EU endpoint) replaces manual log review: it logs every prompt, response, and quality score automatically across all studios, making drift visible before it becomes a problem.

1.5 Key Stakeholders

Stakeholder	Role	Primary interest
Studio owner	Data controller, buyer	Revenue protection, GDPR compliance, minimal overhead. Every member who leaves is someone they know personally.
Coaches	Primary users	Short, useful briefing. No new tool to learn. System meets them in their inbox.
Members	Data subjects	Not feeling surveilled. Clear right to opt out. Data used to support them, not evaluate them.
Dina Bosma	Data processor, operator	Measurable retention outcomes. Replicable system deployable across multiple studios.

1.6 Success Criteria — 90-day POC

Criterion	Measure	Target
Model relevance	Do flagged members churn more than unflagged?	Flagged rate ≥ 2× unflagged (H1)
Intervention outcome	Contacted vs non-contacted flagged members	Contacted members churn less (H2)
Coach adoption	% high-risk alerts acted on within 7 days	> 60%
Briefing quality	Would coaches miss the briefing if it stopped?	≥ 2 of 3 coaches say yes
Reliability	Scheduled runs succeed	≥ 95%
GDPR compliance	No complaints. DPA in place. Deletions on request.	Zero incidents

1.7 Validation Hypotheses

> *H1 - Prediction: Behavioural signals in member visit data reliably predict which members will churn within the next 4–6 weeks. Pass gate (Weeks 1–6): flagged members cancel at ≥ 2× the rate of unflagged members.*

> *H2 - Intervention: When coaches receive AI-generated briefings and reach out, those members retain at a measurably higher rate than at-risk members who were not contacted.*
>
> *Pass gate (Weeks 7–12): contacted cohort retains measurably better.*

H2 is only testable if H1 holds. A failed H1 means recalibrate signals.

A passed H1 with failed H2 means redesign the briefing format: not the model.

Both outcomes are actionable findings.

2. Strategic Plan & Go-To-Market

2.1 Deployment Phases

Phase	Duration	Studios	Goal
Phase 1 - POC	90 days	1	Validate H1 and H2. Prove briefings change coach behaviour. No subscription until both gates pass.
Phase 2 - Pilot	6 months	3-5	Validate across studio types. Activate wearable add-on. Build repeatable onboarding (< 5 days). First subscriptions.
Phase 3 - Scale	Ongoing	10+	Commercial rollout. Parameterised multi-studio workflow. Mindbody/Glofox partnership. ~85% gross margin.

Phase gates: phases move on evidence, not calendar

Gate	All conditions must be met
POC → Phase 2	H1 passes (flagged members cancel ≥ 2× rate) AND H2 passes (contacted members retain measurably better) AND ≥ 2 of 3 coaches say they would miss the briefing.
Phase 2 → Phase 3	New studio onboards in < 5 days. Model accuracy ≥ 80% on real studio data. At least one studio renews beyond pilot. Automated data deletion workflow tested.

> *The monthly retainer includes active model maintenance, including periodic retraining on studio outcome data and monthly drift monitoring, not only system hosting.*

2.2 Go-to-Market

Primary buyer: boutique fitness studio owner-operator in Berlin. Solo owner, 2-5 staff, no data team, €150k–€600k revenue. They feel churn personally; every member who leaves is someone they know.

They know GDPR applies to them but have no lawyer and no time to deal with it.

Not a target: studios where >30% of members use Urban Sports Club or Gympass (different retention economics); large commercial gyms with in-house BI teams; studios with fewer than 80 direct members.

Phase 1- 2 reach: personal introductions in the Berlin fitness network. Studio owner meetups. LinkedIn content on churn data and GDPR tips. The POC studio becomes a referral source once results are visible.

What GDPR fear actually means, and how Spottr removes it.

The studio owner knows GDPR applies to them but has no DPO and no compliance team.

The fear is: "if I add AI and wearables, am I now liable for something I don't understand?" That fear will make them stall and do nothing.

> *"Before we start, we sign a contract. It takes 20 minutes. You update one paragraph in your privacy notice"*
>
> "We use an AI tool to help our coaches understand when members may benefit from additional support. Your coach reviews this information and decides whether to reach out. You can ask us not to use this tool on your data at any time."
>
> *That one sentence can remove the blocker.*

The DPA, consent form, privacy notice wording, and GDPR activity register are all pre-written and included in the (€2,000) setup fee.

The technical controls (consent gate, 48-hour deletion, audit log) mean the studio is demonstrably compliant, not just claiming to be.

2.3 Pricing

Phase	Setup	Subscription	What is included
POC	€2,000 one-time	None during POC	Full setup, 90 days operation, DPA, coach training, debrief with real numbers
Phase 2	€500–1,000	From Day 1	Core briefings + coaching add-on + automated deletion
Small (< 120)	Reduces	€149/month	Core briefings, n8n monitoring, quarterly threshold review
Medium (120–250)	Reduces	€249/month	Core + coaching add-on (check-in - tracking) + automated deletion
Large (250+)	Reduces	€399/month	Full stack including wearable integration + monthly outcome report

2.4 Why Not the Alternatives?

Alternative	What it does	What it misses
Gym software (Mindbody, Gymdesk)	Retrospective attendance reports	No prediction. No coaching context. Shows what happened, not what is about to happen.
Manual process / gut instinct	Works at 30 members	Breaks at 150. Depends on coach memory. Spottr runs whether or not the coach is in the building.
CRM tools (HubSpot, Klaviyo)	Marketing sequences	No fitness-specific signals. Communication feels like marketing which is the wrong tone for a coaching relationship.
Enterprise gym analytics	Multi-location chain tooling	€500–2,000/month+. Requires data teams. Not designed for the boutique coaching relationship model.

> *Spottr is the only system combining churn prediction + coach-language briefings + GDPR compliance built in + a subscription price designed for boutique studios.*

2.5 Commercialisation Model

Phase 1–2: consulting service. Dina owns and operates the pipeline. Studio pays setup + retainer.

Low overhead, fast customisation, builds the case studies needed for Phase 3.

Phase 3: lightweight SaaS transition. Studio onboards via a short form. n8n auto-configures per studio_id. Briefings delivered without per-studio consultant involvement each week.

> *The monthly retainer is a subscription product. Studios do not own the system; they subscribe to its operation. This creates predictable recurring revenue and means the studio is never left maintaining something they cannot support.*

3. ROI & Risk Assessment

3.1 Cost Estimates

Item	Cost	Notes
System setup, config, pipeline, dashboard, compliance docs, coach training (one-time)	€2,000	20–25 consultant hours at €80–100/hr
Infrastructure at pilot scale - n8n free tier, Plotly local, scikit-learn local	€0	No cloud costs at POC scale
OpenAI API (GPT-4o-mini) - 1 batched call/week × ~1,000 tokens × 4.3 weeks/month	~€0.60/mo	92% cheaper than sequential calls
n8n Cloud - shared across studios at pilot scale	~€20/mo	~€4/studio at 5-studio pilot

3.2 Business Value & ROI

Metric	Value	Basis
Revenue at risk	€90,000/year	60% blended churn × 100 members × €1,500 LTV (IHRSA benchmark, upper end)
Conservative case	€11,000/year	40% boutique churn × €1,000 LTV × 20% intervention success
Base case retained revenue	€30,000/year	Retain 20 of 60 at-risk members (33% success) × €1,500 LTV
Coach time savings	€2,600/year	2 hrs/week × 52 weeks × €25/hour
Total annual benefit	**€32,600**	Base case: Retained revenue + time savings
12-month ROI	**1,099%**	Net benefit €29,880 on total year 1 cost of €2,720
Break-even	**33 days**	System pays for itself within the first month of subscription

> *Lead with the conservative case (8x return, €11,000) in the pitch.*
>
> *Use the base case (1,099% ROI) only once POC data validates it.*

3.3 Risk Matrix

Risk	Rating	Mitigation
GDPR non-compliance: wearable data	**L3 × I5 = Red**	Consent gate technically enforces Article 9. Fitbit API blocked if consent_wearable=false. 48-hour deletion. n8n audit log.
Model degradation on real studio data	**L3 × I4 = Red**	Retrain after 6-8 weeks real data. Accuracy must reach ≥ 80%. Fallback to gym_churn_us baseline model.
Data quality gaps from CSV export	**L4 × I3 = Red**	Pre-pipeline data audit. Members with inconsistent data excluded. Confidence score with every classification.
Coach adoption failure	**L3 × I4 = Red**	Email delivery - no new tool. Mandatory 1-hour onboarding. Monthly retrospective. System adjusts on feedback.

Model accuracy degrades over time as studio context changes	**L3 x I4 = Red**	6-monthly retraining on studio outcome data. Monthly churn rate ratio monitoring. Trigger retraining when ratio drops below 1.5.
EU AI Act classification challenge	**L2 × I4 = Amber**	Documented as Limited Risk (Article 50). Human-in-the-loop throughout. n8n audit trail.
LLM hallucination in coaching note	**L2 × I3 = Amber**	Temperature=0. LLM only sees aggregated stats - no individual names in prompt. n8n log reviewed after each run.
USC/Gympass member dilution	**L3 × I3 = Amber**	Studios where >30% of members use USC/Gympass excluded from target market at onboarding screening.

What this table means in plain language:

This is a list of things that could go wrong with the system, how likely they are, and what we have already done to prevent them.
Think of it like a pre-season injury risk assessment; you identify the most likely problems before they happen and put protocols in place.
The rating combines two scores: how likely the problem is to occur (L) and how bad it would be if it did (I).
Red means it needs active controls, amber means we are watching it but it is lower priority.
The most important ones to understand as a coach are these.

First, member data privacy: specifically around Fitbit data.
The system is built so that wearable data is technically impossible to access unless a member has signed a consent form. It is not a policy that someone could forget to follow; it is a lock in the software.
Second, the model may not be perfectly accurate on your studio's real members straight away, because it was trained on data from a different gym.
That is expected and planned for: after 6 to 8 weeks of real data from your studio, it gets retrained and improves.
Third, if your studio has a lot of members who come via Urban Sports Club or Gympass rather than direct membership, those members behave differently when it comes to cancellation, so the system is not designed for them and they are filtered out.
And *finally*, the biggest operational risk is simply coaches not reading the Monday email, which is why the system was designed to land in your inbox with no new app, no new login, and no extra steps.
You open an email. That is it.

4. MVP Documentation

4.1 MVP Scope

The MVP demonstrates two things end-to-end:

(1) core system: visit data → churn risk scores → weekly email report for all members;

(2) coaching add-on: check-in + optional Fitbit wearable data → coaching-specific briefing.

Everything runs on synthetic data for the demo. The architecture mirrors a real studio deployment exactly.

4.2 Data Inputs

File	Role	Contents
gym_churn_us.csv	Model training only	4,000 labelled gym members with real churn outcomes. Used once to train churn_model.pkl. Never shown in the dashboard.
demo_members.csv	Demo population	150 members derived from 167,783 Endomondo sessions. 60 HIGH / 30 MEDIUM / 60 LOW. Replaced with real studio CSV at go-live.
coaching_sessions.csv	Coaching add-on	Synthetic coaching attendance: member_id, session_date, session_booked (bool), session_attended (bool), coach_id
coaching_consent_flags.csv	Consent tracking	member_id, checkin_consent, general_attendance_consent, wearable_consent (all boolean)

4.3 Model Files

File	What it does	When to run
train_churn_model.py	Reads gym_churn_us.csv, trains logistic regression (StandardScaler + LogisticRegression, max_iter=1000), evaluates on 20% hold-out set, saves churn_model.pkl. Already run — model exists.	Once. Or when new labelled data is available.
score_members.py	Reads demo_members.csv, runs every member through churn_model.pkl, writes scored_members.csv with churn probability and HIGH/MEDIUM/LOW tier per member.	Weekly before Monday trigger. Or on demand for a fresh scoring.
scoring_api.py	Local web server. n8n can POST one member's data to localhost:5000/score and receive {churn_probability, churn_risk_tier} in real time. Demonstrates the production architecture.	Manually for demos. In production: deployed to Railway so n8n calls it over the internet instead of localhost.

Current state: churn_model.py lives only on my laptop.

The scoring API runs only when manually started. For the POC this is fine: I ran score_members.py before the demo and the n8n workflow read the output CSV. For go-to-market, scoring_api.py deploys to Railway so n8n calls it over the internet.

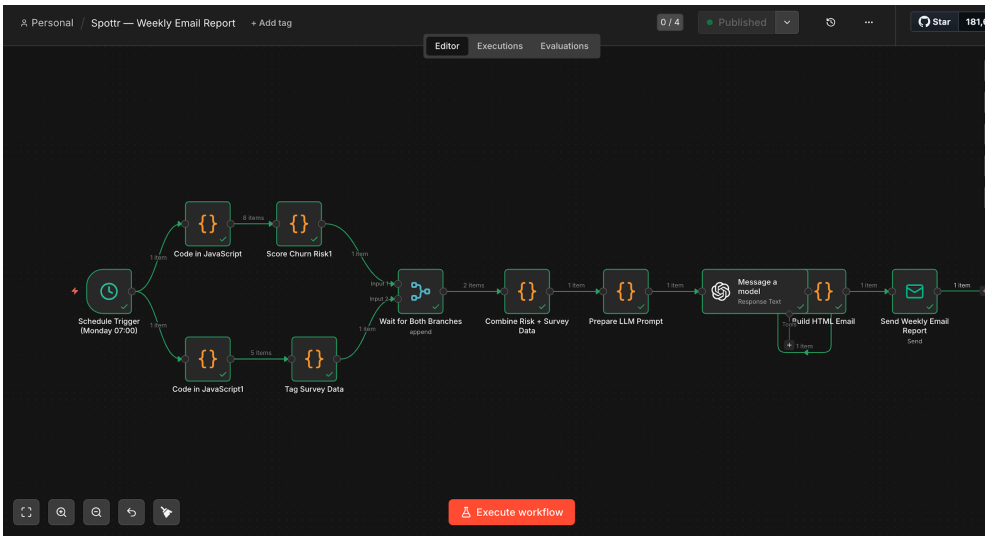
4.4 n8n Workflow Architecture

Demo walkthrough: https://github.com/dlbosma-hue/bootcamp_env/issues/2#issue-4186433120

Pipeline 1: Weekly email (Monday 07:00)

Schedule trigger fires → two parallel branches run simultaneously: (A) load member data → score churn risk → filter HIGH + MEDIUM; (B) load survey responses from past 7 days → tag WATCH members (LOW risk but satisfaction ≤ 2).

Both branches merge → single GPT-4o-mini batched prompt generates all coaching notes → build 3-section HTML email → SMTP send to all coaches and owner.



Spottr Weekly Email Report workflow in n8n — all nodes green, published and live.

Pipeline 2: Survey trigger (daily, event-based)

Trigger A (20:00): checks each class for attendance drop >30% vs 4-week average for 2+ consecutive sessions.

Trigger B (22:00): checks total studio check-ins vs 7-day average (>20% drop). If threshold breached → load cooldown log → filter members not surveyed in past 5 weeks → send Tally.so survey email → log send.

Tally webhook fires on every submission → parse and log response to survey_responses.csv.

Monthly Survey Summary trigger (1st of each month, 08:00): reads all responses from the previous calendar month from survey_responses.csv → aggregates total responses, average satisfaction, average NPS, count of members rated ≤ 2/5, goal progress breakdown, and all feedback quotes → sends formatted email to studio owner via SMTP.

Pipeline 3: Coaching add-on (Monday 07:00): load coaching clients → split per member → Consent Gate (IF node) → TRUE: call Fitbit API and merge wearable data / FALSE: use check-in data only → merge → score coaching risk → GPT-4o-mini briefing → send coaching email.

Consent gate

The n8n coaching workflow checks the consent_wearable boolean before any Fitbit API call. If false, the Fitbit branch is skipped and the workflow sends a check-in-only briefing. This is a technical enforcement of GDPR Article 9, not a policy control.

4.5 Data Ingestion by Phase

Phase	Who exports	Method	Frequency	Notes
Phase 1	Studio staff	Manual CSV export from gym management system, dropped into pipeline folder	Weekly before Monday 07:00	3 columns needed. Other 10 filled from training medians.
Phase 2A	Nobody	Scheduled email: Mindbody/Glofox	Scheduled (e.g. Sunday night)	No studio action after initial setup.

		auto-sends CSV by email; n8n receives and saves		
Phase 2B	Nobody	Direct API: n8n calls Mindbody/Glofox API directly: live data, no file	Monday at trigger time	Requires API credentials from studio.
Phase 3	Nobody	Webhook: gym system pushes event on every check-in: near real-time scoring	Real-time	Also enables automated deletion on membership cancellation.

4.6 Output Formats - Live Screenshots

Weekly email:

Spottr Weekly Report

Friday, 27 March 2026

8	6	4	1
Total Members	At-Risk	High Risk	Watch

🚨 Members Needing Attention

Sarah K. HIGH

📅 25 days since last visit | 📊 0.8 sessions/week | 📅 14 months tenure

💬 "Been busy with work, hoping to get back on track soon." — 3/5 satisfaction

> **Coaching note:** This member has not visited in 25 days and expressed feeling busy with work. Despite a long tenure, their attendance is below average. A conversation about creating a flexible plan to ease back into routine might resonate.

Emma V. HIGH

📅 30 days since last visit | 📊 0.5 sessions/week | 📅 9 months tenure

💬 "Feeling unmotivated, not sure the plan suits me anymore." — 2/5 satisfaction

> **Coaching note:** This member has been absent for 30 days and feels unmotivated, rating their experience low. Exploring their reasons for feeling disconnected and discussing alternative plans could help reignite their engagement.

David L. HIGH

📅 22 days since last visit | 📊 0.7 sessions/week | 📅 11 months tenure

💬 "Schedule conflicts making it hard to come in regularly." — 2/5 satisfaction

> **Coaching note:** With 22 days since last visit and schedule conflicts reported, this member's attendance is suffering. Initiating a dialogue about finding convenient times or classes may assist in enhancing their participation.

Members Needing Attention: HIGH risk cards with days inactive, sessions/week, survey quote, and coaching note.

👥 Satisfaction Alerts

These members are **still attending** but rated satisfaction 2/5 or lower. No urgency — a warm personal check-in this week can prevent this becoming an attendance problem.

Lisa M. WATCH

Attending normally — 3.5 sessions/week | last visit 3 days ago

Satisfaction: 2/5 | NPS: 4/10

💬 "Classes are good but I feel like my progress has stalled."

> **Coaching note:** Despite attending regularly, this member reports low satisfaction. A curious and supportive dialogue focused on their progress and finding ways to enhance their experience could help address their feelings of stagnation.

📋 Survey Snapshot (Last 7 Days)

Responses received: **5**
Avg satisfaction: **2.6/5**
Avg NPS: **4.6/10**

Generated by Spottr · Monday, 30 March 2026

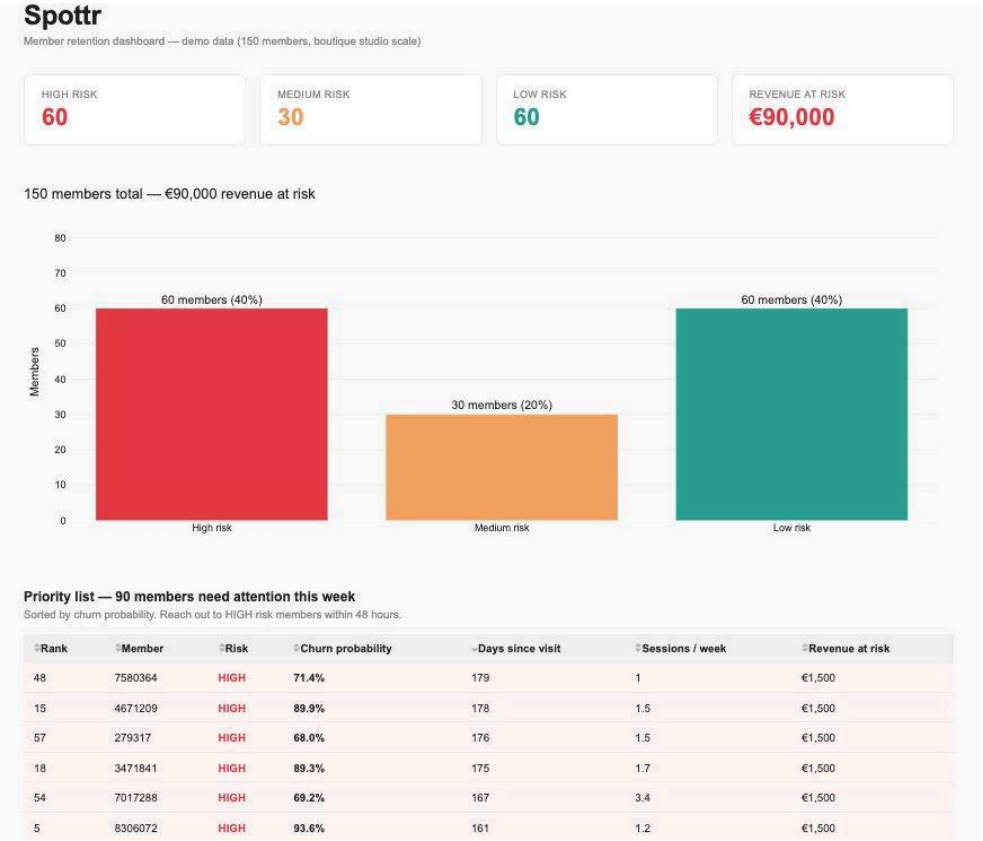
Satisfaction Alerts (WATCH flag) and Survey Snapshot. WATCH members are still attending but rated satisfaction ≤ 2.

The WATCH flag

A WATCH member is not flagged by the churn model, they are still attending normally. But they rated satisfaction 2/5 or lower in a survey. They are present but unhappy. The system catches this separately and shows these members in a purple section of the email with a different coaching tone: warm and curious, not urgent.

> *"A member can be showing up every week and still be about to leave; because they're unhappy, not because they've gone quiet. The WATCH flag catches that before it becomes an attendance problem."*

Retention dashboard



Rank	Member	Risk	Churn probability	Days since visit	Sessions / week	Revenue at risk
48	7580364	HIGH	71.4%	179	1	€1,500
15	4671209	HIGH	89.9%	178	1.5	€1,500
57	279317	HIGH	68.0%	176	1.5	€1,500
18	3471841	HIGH	89.3%	175	1.7	€1,500
54	7017288	HIGH	69.2%	167	3.4	€1,500
5	8306072	HIGH	93.6%	161	1.2	€1,500

Spottr retention dashboard: demo data, 150 members. Colour-coded risk tiers. Sortable priority table with churn probability, days since visit, and revenue at risk.

The dashboard is not the headline, the weekly email is.

The dashboard answers a different question: when the owner asks 'how bad is it really?' it answers in 5 seconds.

Phase 1: runs at localhost:8050 on studio computer only.

Phase 2: hosted on Railway (~€5–7/month extra): accessible from any device via URL.

Coaching add-on email (31 March 2026 - live, real Fitbit + GPT-4o-mini)

Coaching Update — Tuesday 31 March

High Priority

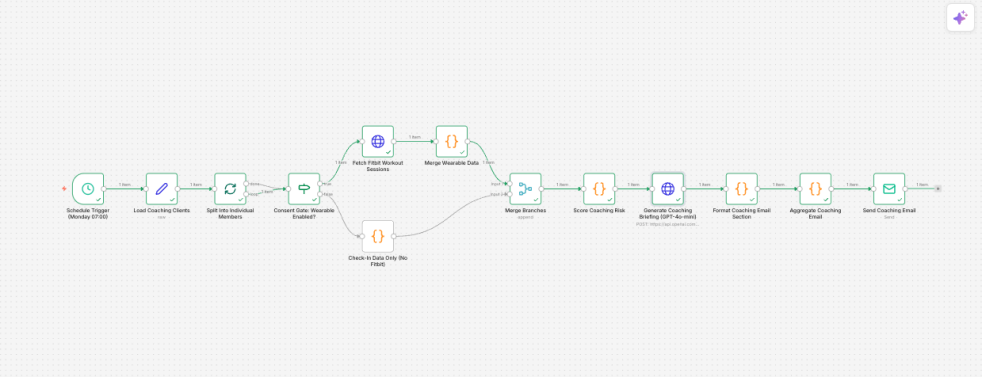
M001 — HIGH PRIORITY
Member M001 has attended 3 out of 8 sessions, giving them an attendance rate of 38%. It's been 16 days since their last check-in, and they haven't engaged in any wearable sessions in the past two weeks. This might be a good opportunity to reconnect and see how they're feeling about their progress and any challenges they might be facing.
Data source: checkin_and_wearable

This briefing is generated automatically from coaching session attendance and wearable data (where consented). Member names are not processed by the AI system. For questions or to update consent settings, contact your system administrator.

This email was sent automatically with n8n

Coaching Update email: M001 flagged HIGH PRIORITY: 38% attendance, 16 days since check-in, 0 wearable sessions in 14 days.

Data source: checkin_and_wearable.



Wearable workflow: all nodes green. Consent Gate routes TRUE (Fitbit API) vs FALSE (check-in only).

Wearable workflow: live execution result

Node	Result
Load Coaching Clients	1 item - Sophie Müller (M001)
Fetch Fitbit Workout Sessions	Real Fitbit API call; authenticated. activities: 0 (no sessions in last 14 days for test user)
Merge Wearable Data	1 item: wearable_sessions_14d: 0
Score Coaching Risk	1 item: HIGH risk (37.5% attendance rate, 16 days since last check-in)
Generate Coaching Briefing	Real GPT-4o-mini call: coaching briefing generated live
Format → Aggregate → Send	Email dispatched to coach@gmail.com: Coaching Update 31 March confirmed in inbox

Class dropped: Survey sent to attendees (5 weeks cool-off)

Hi,

We noticed things have been a little quieter at the studio lately and we wanted to check in with you. It would really help us to hear how things have been going for you.

We have a few quick questions, it will take less than a minute:

Share your feedback

You can also just reply to this email if you prefer to write directly.

Thanks,
The team at YOUR GYM

You can opt out of these check-in emails at any time by replying with 'unsubscribe'. Your response is voluntary and confidential.

This email was sent automatically with n8n

Survey request per email (less than 1 minute) to attendees of a lower-as-usual visited class.

Monthly survey summary (1st of the month, sent to the owner only)

📊 Spottr — Monthly Survey Summary

March 2026

6	2.6	4.6	3
Responses	Avg Satisfaction (1–5)	Avg NPS (0–10)	Low Satisfaction Alerts

⚠️ **3 members rated satisfaction ≤ 2** — consider proactive outreach.
• M003 (score: 2) • M005 (score: 2) • M008 (score: 2)

📋 All Responses

Member	Date	Satisfaction	NPS	Goal Progress	Feedback
UNKNOWN	2026-03-31	N/A	N/A	-	-
M001	2026-03-24	3	5	falling_behind	Been busy with work, hoping to get back on track soon.
M003	2026-03-25	2	4	falling_behind	Classes are good but I feel like my progress has stalled.
M005	2026-03-21	2	3	falling_behind	Feeling unmotivated, not sure the plan suits me anymore.
M007	2026-03-26	4	7	getting_started	Still finding my rhythm as a new member.
M008	2026-03-23	2	4	falling_behind	Schedule conflicts making it hard to come in regularly.

🎯 Goal Progress Breakdown

- **not specified:** 1 member
- **falling behind:** 4 members
- **getting started:** 1 member

Generated by Spottr Automation · 31 March 2026

Monthly Survey Summary: March 2026. 6 responses, avg satisfaction 2.6/5, avg NPS 4.6/10, 3 low-satisfaction alerts. Full response table with member ID, scores, and feedback.

4.7 Multi-Studio Workflow Scaling

Phase	Architecture
POC	One n8n workflow copy per studio. Hardcoded email addresses and data file path. ~30 minutes to configure a new studio. Correct for 1-3 studios.
Phase 2	One parameterised workflow reads studio_id at startup and looks up config from studios_config.csv (coach emails, data path, thresholds). Separate trigger per studio, same workflow.
Phase 3	Scoring API deployed to Railway. n8n calls via HTTP with studio_id in payload. Studio data in database rows, not local CSVs. Adding a studio = adding a row and activating one trigger.

5. Proof of Concept

5.1 What the POC Is

A structured 90-day engagement with one boutique studio to answer two questions in sequence. Not a trial.

A paid engagement (€2,000) with a clear outcome review at the end.

5.2 The Honest Pitch - What I Am Selling

The model is validated: 92.5% accuracy and 0.977 ROC-AUC on 4,000 real gym member records.

What is not yet proven is whether coaches acting on the briefings actually retain members.

That is the human outcome, and it requires the POC to measure.

> *The ask is not 'buy a subscription.' It is 'fund the proof.' €2,000 to find out whether this saves your studio €30,000 a year should not be a hard sell.*

Objection	Answer
"What if it doesn't work?"	Then you've spent €2,000 finding that out before committing to anything monthly. I'd rather prove it works than ask you to trust it.
"What about GDPR?"	Everything is pre-written. We sign a contract before anything starts. You update one sentence in your privacy notice; I'll write it for you.
"How much work for my coaches?"	Monday morning: read a three-minute email. That's it. They decide who to reach out to, in their own words.

5.3 Two-Stage Validation

Stage 1: Model validation (Weeks 1–6)

The system runs without coaching interference to establish a baseline.

At Week 6: do flagged members cancel at ≥ 2× the rate of unflagged members?

If yes → proceed to Stage 2. If no → recalibrate thresholds.

Do not proceed to Stage 2 until Stage 1 passes.

How the Stage 1 comparison is done in practice:

At the start of Week 1, Spottr produces its first risk list. Every active member receives a label: HIGH, MEDIUM, or LOW, based on their visit history. That list is frozen. Coaches receive their Monday briefings as normal during Weeks 1-6, but do not act on them yet.

This is deliberate: if coaches start reaching out to flagged members during the observation window, we can no longer tell whether a member stayed because the model was right or because the outreach saved them. The observation period needs to be clean.

At the end of Week 6, the studio owner exports a cancellation list from their gym management system, this is a standard report available in Mindbody, Glofox, Gymdesk, and most booking tools, or it can be a simple spreadsheet the front desk updates. We then do one calculation: what percentage of HIGH-risk flagged members cancelled, and what percentage of unflagged members cancelled? Dividing the first number by the second gives the churn rate ratio. A ratio of 2 or above means the model is identifying the right people. A ratio below 2 means the model needs recalibration before coaches start acting on it.

One practical caveat: if fewer than five members cancel in total during the six-week window, the sample is too small for the comparison to be conclusive. This can happen at studios with very low monthly churn or seasonal patterns. In that case the Stage 1 window is extended by four weeks and the same calculation is repeated. This possibility should be discussed with the studio owner at setup and noted in the Week 1 configuration session.

Stage 2: Intervention validation (Weeks 7–12)

Coaches begin acting on briefings.

Outreach tracked per flagged member. At Week 12: do contacted flagged members retain at a measurably higher rate than non-contacted flagged members?

If yes → Phase 2 begins. If no → redesign briefing format. The model is working; the intervention needs work.

5.4 Outcome Matrix

H1	H2	Meaning	Next step
Pass	**Pass**	Full loop validated	Proceed to Phase 2. Begin subscription.
Pass	**Fail**	Model accurate, intervention broken	Redesign briefing format and outreach protocol before scaling.
Fail	**N/A**	Model not accurate on real data	Recalibrate signal thresholds. Retrain on studio data. Retest H1.

5.5 Measurement Framework

Stage 1 metrics (Weeks 1-6)

Metric	How measured	Frequency
Members flagged HIGH / MEDIUM / LOW	System log	Daily
Actual cancellations: flagged cohort	Studio membership records	Weekly
Actual cancellations: unflagged cohort	Studio membership records	Weekly
Churn rate ratio (flagged vs unflagged)	Calculated from above	End of Week 6

Stage 2 metrics (Weeks 7-12)

Metric	How measured	Frequency
Flagged members contacted within 7 days	Coach self-report or CRM log	Weekly
Churn rate: contacted flagged members	Studio outcome data	End of Week 12
Churn rate: non-contacted flagged members	Studio outcome data	End of Week 12

Coach satisfaction with weekly email	Short 3-question survey	Monthly
Average member satisfaction score (monthly)	Monthly survey summary email	Monthly - 1st of each month
% members rated satisfaction ≤ 2/5	Monthly survey summary email	Monthly - 1st of each month

5.6 What the Studio Gets in the POC

- Weekly Monday email: which members are going quiet, why, and coaching context per member.
- Retention dashboard on studio computer: full member risk view, colour-coded, sortable.
- Member satisfaction survey: auto-triggered on attendance drops, 5-week cooldown per member.
- WATCH flag: members still attending but satisfaction ≤ 2, surfaced separately with different coaching tone.
- Monthly survey summary email: sent to studio owner on the 1st of each month; aggregated satisfaction scores, NPS, goal progress breakdown, and every feedback quote from that month.
- All GDPR documentation pre-written and ready to sign (DPA, consent form, privacy notice wording).
- 90-day debrief with real outcome data: the 18x ROI argument only gets made here, with real numbers.
- NOT in POC: wearable integration, automated deletion, multi-studio infrastructure, online dashboard.

5.7 Technical Specifications

Tools Used and Why

Tool	Role	Why chosen
scikit-learn (Python)	Logistic regression churn model	Lightweight, interpretable, runs locally - no cloud compute needed for a 150-member studio
pandas	Data ingestion and feature engineering	Standard library for CSV processing; handles the member visit data pipeline
n8n	Workflow automation - weekly email, survey trigger, monthly summary	No-code/low-code; all three workflows are exportable JSON; runs on a schedule without manual input
Plotly Dash (Python)	Retention dashboard	Browser-based, no login required, bookmarkable URL; studio computer opens it like any website
OpenAI GPT-4o-mini	Coaching note generation	Cost-effective (approx. €0.10–0.20 per weekly studio run); consistent tone and format
Tally	Member satisfaction survey form	Free, no-login-required form; webhook output writes directly to survey_responses.csv
SMTP (Gmail/Outlook)	Email delivery — weekly briefing and monthly summary	Studio already has an email account; no additional service required

What the POC Does: Step by Step

Step 1 - Data ingestion

The studio exports member visit data from their booking system (Mindbody, Glofox, Gymdesk, or a spreadsheet). The file is loaded with pseudonymous member IDs. No names enter the pipeline.

Step 2 - Churn scoring

A logistic regression model — trained on 4,000 labelled gym members — scores each active member. Output per member: risk tier (HIGH / MEDIUM / LOW / WATCH), churn probability (0–1), and revenue at risk in euros.

Step 3 - Coaching note generation

For each HIGH and MEDIUM-risk member, GPT-4o-mini writes one plain-English paragraph summarising the attendance pattern and suggesting a coaching angle. The AI does not contact members. The note goes to the coach only.

Step 4 - Weekly email briefing (Monday 07:00)
The n8n workflow triggers every Monday at 07:00, reads the latest scored output, formats the briefing email (members ranked by risk, coaching notes, WATCH flags, any survey responses from that week), and sends it to the coaching team via SMTP.

Step 5 - Event-triggered member survey
A separate n8n workflow monitors class attendance. When a class records a drop in attendees, it sends a 3-question Tally survey link by email to members who attend that class. Responses are written to survey_responses.csv via webhook.

Step 6 - Monthly summary (1st of month, 08:00)
The survey workflow runs a separate trigger on the 1st of every month. It aggregates all survey responses from that calendar month and sends a summary to the studio owner: satisfaction scores, NPS trend, WATCH candidate count, and every free-text quote.

Step 7 Retention dashboard
A Plotly Dash app reads the scored output and renders a live browser dashboard: all members colour-coded by risk tier, total revenue at risk, and a sortable priority table. Refreshes automatically before each Monday briefing.

Known Limitations - POC vs. Production

Limitation	POC behaviour	Production fix
Manual data ingestion	Studio exports CSV manually and shares with consultant	API integration with Mindbody / Glofox / Gymdesk booking systems
Single-studio model	Logistic regression trained on generic dataset, calibrated per studio	Per-studio retraining with accumulated outcome data after 2–3 POC completions
No automated deletion	Manual deletion on request within 48 hours	Automated scheduled deletion job with audit log
n8n self-hosted	Runs on consultant's local machine or a single VPS	Cloud-hosted n8n instance with uptime monitoring per studio
Survey delivery via email only	Tally link sent by email	Native SMS or WhatsApp delivery option for higher response rates
Dashboard not authenticated	Anyone with the URL can view the dashboard	Auth layer (simple password or OAuth) before multi-studio rollout
GPT-4o-mini hallucination risk	Coaching notes reviewed manually by consultant before go-live	Structured output validation + automated tone/length checks

5.8 Model Drift and Retraining

The churn model was trained once. Over time its accuracy will decrease for two reasons: the studio changes (new instructors, pricing shifts, member demographics), and external conditions change (seasonality, local competition, broader disruptions). This is called concept drift. The relationship between the input signals and churn outcomes shifts gradually, and a model that is never retrained will start flagging the wrong people and erode coach trust in the briefing.

The monitoring metric is the churn rate ratio from Stage 1: flagged members cancelling at twice the rate of unflagged members. This should be tracked monthly after the POC ends, not only at the Week 6 gate.

A ratio between 1.5 and 2.0 is an early warning. A ratio below 1.5 triggers retraining.

Ratio	What it means
2.0 or above	Model is performing well. No action needed.
1.5 to 2.0	Early warning. Signal is weakening. Review threshold settings.
Below 1.5	Drift confirmed. Retrain the model on accumulated studio outcome data before the next briefing cycle.

Retraining means running the training script on real cancellation outcomes accumulated from the studio, not rebuilding from scratch. Minimum data required is 6 to 8 weeks of real check-in data with known outcomes.

The retrained model must reach at least 80% accuracy on a held-out test set before replacing the current version. If it does not, the previous model continues to run.

- Suggested schedule: every 6 months per studio, or immediately when the ratio drops below 1.5.
- This is a standard part of the monthly retainer. Not just hosting, but active model maintenance. It is what justifies ongoing consultant involvement beyond the POC.

6. Coach Quick-Start Guide

> *Spottr watches your members' data automatically. Every Monday at 07:00, you get one email. You read it, you decide who to reach out to. The system does not contact members: you do.*

6.1 Your Weekly Email

Subject: Spottr Weekly Retention Report. Three sections: (1) HIGH and MEDIUM risk members — days since last visit, sessions/week vs historical norm, coaching note; (2) WATCH members — still attending but rated satisfaction ≤ 2; (3) survey responses from the past week.

Tier	What it means	Action
HIGH	Model probability ≥ 65%. Strong disengagement pattern.	Reach out within 48 hours.
MEDIUM	Model probability 35–65%. Early signal, not yet critical.	This week, lower urgency.
WATCH	Still attending but satisfaction ≤ 2 in the survey. Sentiment signal, not churn signal.	Warm check-in this week.
LOW	Attending normally. No action needed.	No action needed.

6.2 The Dashboard

Run 'python dashboard/spottr_dashboard.py' in the terminal, then open http://127.0.0.1:8050. Keep the terminal open while using it.

The dashboard is not the headline, the weekly email is. The dashboard answers a different question: when the owner asks 'how bad is it really?' it answers in 5 seconds.

To pitch as: 'Your coaches get a briefing every Monday. And if you want to see the full picture, all 150 members, colour-coded, ranked, with revenue at stake, it is one click.'

Phase 1: localhost only. Phase 2: hosted on Railway (~€5–7/month extra), accessible from any device via URL.

6.3 Enrolling a Member for Wearable Tracking

Optional. For private coaching clients only.

Step 1: have the conversation: explain what Spottr collects and why, only proceed with a clear yes.

Step 2: member signs the consent form.

Step 3: record consent in data/fitbit_tokens.json (set consent_wearable: true, consent_tier: 'workout_only').

Step 4: run python src/data_loaders/fitbit_auth.py --member [ID]: member approves in browser.

Step 5: verify next Monday's email footer confirms correct data sources.

6.4 Withdrawing Wearable Consent

> *All five steps must be completed within 48 hours. GDPR requirement: not optional!*

- Note date and time of request. No reason required from the member.
- Set consent_wearable: false in data/fitbit_tokens.json. Add consent_withdrawn date. Save. n8n stops fetching on the next run.
- Delete workout data: python src/data_loaders/fitbit_loader.py --delete [member_ID]
- Revoke Fitbit token: Fitbit app → Account → Manage Data → Connected Apps → Spottr → Remove
- Send written confirmation to members within 48 hours. Keep a copy. Required by the DPA.

6.5 The Member Survey

Triggered when class attendance drops >30% across 2 consecutive sessions, or studio check-ins drop >20% vs 7-day average. Max once per member per 5 weeks. Three questions: training rating 1- 5, barriers (optional free text), coach check-in request (yes/no).

If a member says yes - contact them the same week. Responses appear automatically in Monday's email.

6.6 The Monthly Summary

On the 1st of each month, the studio owner receives a consolidated summary of all survey responses: total responses, avg satisfaction, avg NPS, members who rated ≤ 2 flagged for outreach, full response table.

Use it in monthly conversations with your Spottr consultant to track whether sentiment is improving.

6.7 Quick Reference

Task	How
Read weekly coach briefing	Email - Monday 07:00
Read monthly owner summary	Email - 1st of each month
Open dashboard	python dashboard/spottr_dashboard.py → http://127.0.0.1:8050
Enrol member for wearable	fitbit_tokens.json → python fitbit_auth.py --member [ID]
Member withdraws wearable consent	5-step process within 48 hours - see Section 6.4
WATCH member in email	Warm check-in this week. Still attending: sentiment signal, not churn.
Something looks wrong	Contact studio manager or Dina (consultant)

7. EU AI Act Compliance

7.1 Classification - LIMITED RISK (Article 50)

Element	Decision
Risk classification	**LIMITED RISK - Article 50 (Transparency Obligations only)**
Conformity assessment required?	No - no notified body review required.
High-risk (Annex III)?	No - not employment decisions, not essential services, not biometric categorisation.
Unacceptable risk (Article 5)?	No - no subliminal manipulation, no exploitation, no biometric surveillance, no social scoring.

7.2 Article 50 Obligations - How They Are Met

Obligation	Implementation
Transparency to system users	Every weekly email shows the behavioural triggers per member (days inactive, session frequency drop). Coaches understand what was flagged and why.
Information to data subjects	Privacy notice states AI tool analyses attendance. Consent form lists wearable data used and why. The right to object is explicit and honoured within 24 hours.
Human oversight - no automation	The system generates recommendations only. Coaches make all contact decisions. No automated member notifications or membership changes.
Accuracy and robustness	92.5% accuracy, 0.977 ROC-AUC on 4,000 labelled members. Members with < 4 weeks data excluded. Confidence score accompanies every classification.
Audit trail	n8n execution log records every prompt, AI output, and timestamp. DPA assigns controller/processor roles. Article 30 register maintained.

8. GDPR Documentation

8.1 Data Flow Map

Data type	Source	Identifier	Where it goes	Deletion
Member visit history	Gym CSV export	Member ID (pseudonym)	Pipeline → model → GPT-4o-mini → email	30 days after membership end
Wearable activity data	Fitbit API (consent-gated)	Member ID (pseudonym)	Consent gate → local script only. Never sent to OpenAI.	48 hrs after consent withdrawal
Survey responses	Tally.so → n8n webhook	Member ID (pseudonym)	survey_responses.csv → weekly email	90-day rolling
n8n execution logs	n8n Cloud EU	None (operational)	n8n Cloud EU region	3 months (n8n default)

Key principles: pseudonymisation (member names never enter pipeline); data minimisation (only churn prediction fields extracted); consent gating (wearable blocked at workflow level until consent confirmed); EU data residency (n8n EU region; OpenAI receives only aggregated non-personal prompt data, SCCs apply).

8.2 Processing Register (Article 30)

Activity	Purpose	Legal basis	Retention	Sub-processors
1 - Churn scoring	Identify at-risk members	Legitimate interest Art. 6(1)(f)	Membership + 30 days	OpenAI (SCCs), n8n (EU)
2 - Wearable data	Coaching add-on signals	Explicit consent Art. 9(2)(a)	48 hrs after withdrawal	Fitbit (SCCs)
3 - AI output log	Quality audit trail	Legitimate interest Art. 6(1)(f)	3 months	n8n (EU)
4 - Survey responses	Member feedback	Legitimate interest Art. 6(1)(f)	90 days	Tally.so (EU), n8n (EU)

8.3 DPIA Summary - Wearable Data

Wearable data (in-session heart rate, session metadata) qualifies as health data under Article 9 GDPR.

It is not necessary for the core service - the model achieves 92.5% accuracy on visit history alone.

It is offered as an optional coaching add-on only, under separate tiered explicit consent (3a: workout sessions; 3b: full daily activity including resting HR).

The n8n consent gate technically enforces this before any Fitbit API call.

9. Data Processing Agreement

Must be signed before any member data is processed. Template for use at every studio deployment.

Party	Role	Identity
[STUDIO NAME]	**Controller**	Registered business at [address]. Determines purpose and means of processing member data.
Dina Bosma-Buczynska	**Processor**	Independent AI systems consultant. Processes data only on behalf of and under instruction from the Controller.

Key provisions

- The processor processes personal data only on documented instructions from the Controller.
- The processor shall not use member data for any purpose other than delivering the Spottr system.
- Pseudonymisation applied before data enters the AI pipeline. API keys never committed to version control.
- Consent gate technically enforces Article 9: wearable data never accessed without explicit consent flag set.
- Erasure requests: completed within 48 hours of written instruction. Written confirmation sent to the Controller. Deletion log maintained.
- Breach notification to Controller within 24 hours. The controller notifies supervisory authority (Berliner Beauftragte) within 72 hours.

Subprocessors

Subprocessor	Purpose	Location	Mechanism
OpenAI	GPT-4o-mini LLM inference - coaching briefings	USA	SCCs
n8n Cloud	Workflow automation, scheduling, email delivery	EU region	EU-hosted
Fitbit	Wearable OAuth token exchange (coaching add-on only)	USA	SCCs
Tally.so	Survey form delivery and webhook response logging	EU	EU-hosted
Studio SMTP	Weekly briefing email delivery	Per studio	EU preferred

Controller signature	Processor signature
Name: _____	Name: Dina Bosma-Buczynska
Date: _____	Date: _____

10. Green Tech Assessment

Prepared by Dina Bosma-Buczynska · March 2026 · Design-time audit - assessed before live deployment.

> *Functional unit (R): one weekly retention briefing delivered to one studio — churn risk scores for all members and coaching context per flagged member. All estimates are per R.*

10.1 System Footprint

Component	Activity per R	Footprint
scikit-learn (local)	Logistic regression on 150 rows - sub-second on CPU, no GPU required	Negligible. No cloud inference.
GPT-4o-mini (batched)	1 API call per week. ~1,000 input + ~400 output tokens. Batched - not per member.	Moderate. Dominant energy driver. US-hosted (~0.4 kgCO2/kWh).
n8n workflow	1 scheduled execution per week (52/year). Event-based survey trigger only fires on real signal.	Low. Weekly cadence aligned with service value.
Plotly dashboard	Loads CSV on each browser refresh. On-demand only.	Negligible at single-studio scale.

10.2 Green Decisions Already Implemented

Decision	Alternative rejected	Energy saving
One batched GPT-4o-mini call per run	One call per flagged member (N sequential)	~80% token reduction at 10-member list
Weekly trigger (52×/year)	Daily trigger (365×/year)	86% reduction in n8n executions
Local logistic regression (CPU)	Cloud-hosted ML inference (GPU)	Eliminates cloud inference entirely at POC scale
GPT-4o-mini (small, fast model)	GPT-4o or larger model	Lower per-token energy footprint

10.3 Improvement Roadmap

Priority	Phase	Action	Expected outcome
1 - Token logging	**Phase 1**	Read usage.total_tokens from OpenAI API response in n8n after each run. Append to token_log.csv. 30-minute implementation.	Concrete measurable energy proxy from first live run.
2 - EU LLM option	**Phase 2**	Evaluate Mistral (EU-hosted) as drop-in replacement for GPT-4o-mini. Compare quality on sample briefing set.	~50% carbon reduction per call (EU vs US grid).
3 - Dashboard cache	**Phase 2**	Cache CSV on startup. Refresh only when a new briefing CSV is written.	Eliminates redundant file I/O at multi-studio scale.

11. Coach & Wearable Quick Reference

Consolidated coach-facing operational materials for printing or sharing with studio staff.

11.1 What Data Spottr Uses

Data type	Required?	Legal basis	What it enables
Coaching session check-in	**Required**	Legitimate interest	Attendance rate, missed sessions, days since last check-in
General studio attendance	**Optional**	Legitimate interest	Distinguishes full disengagement from coaching-only drop-off
Wearable workout sessions (3a)	**Optional**	Art. 9(2)(a) consent	Session type, duration, calories, in-session HR avg/peak
Full daily activity (3b)	**Optional**	Art. 9(2)(a) consent (needs 3a)	Daily steps, active minutes, resting heart rate

> *Never collected: resting HR (without 3b consent), sleep data, GPS location, continuous passive monitoring, medical history, financial data.*

11.2 Enrolment Checklist

- Have the conversation - explain what Spottr collects and why. Only proceed with a clear yes.
- Consent form signed - give members the form, they sign, keep a copy.
- Record in data/fitbit_tokens.json - consent_wearable: true, consent_tier, consent_date, consent_signed: true.
- Run Fitbit OAuth - python src/data_loaders/fitbit_auth.py --member [ID] — member approves in browser.
- Verify next Monday's email footer matches what member consented to.

11.3 Withdrawal Checklist (within 48 hours)

- Note date and time. No reason required.
- Set consent_wearable: false in fitbit_tokens.json. Add consent_withdrawn date. Save.
- Delete workout data: python src/data_loaders/fitbit_loader.py --delete [member_ID].
- Revoke Fitbit token: Fitbit app → Account → Manage Data → Connected Apps → Spottr → Remove.
- Send written confirmation to members within 48 hours. Keep a copy.

> *After withdrawal: member still appears in weekly email if check-in data triggers a risk flag. Check-in history continues under legitimate interest — separate from wearable consent.*

11.4 Member Data Rights

Right	What to do
Access	Export their rows from all CSV files. Deliver within 30 days.
Erasure	5-step deletion process within 48 hours. Written confirmation required.
Rectification	Correct in source system. Pipeline re-run within 48 hours.
Objection	Add exclude_from_scoring: true to their entry. Inform consultant. Confirm within 24 hours.
Withdraw consent	Follow the withdrawal checklist (Section 11.3) within 48 hours.

11.5 What to Say to Members

> *Do not tell a member they were flagged by an AI system. Reach out naturally.*
>
> *If a member asks: "It is a system we use to keep track of how our members are doing. It helps me make sure I do not miss anyone who might benefit from a check-in. You can ask us not to use it on your data at any time."*

Spottr

Predicts who is about to leave your studio and makes sure your coaches know before it is too late.

The prediction model is validated: 92.5% accurate on 4,000 real gym member records.

Whether coaches acting on it actually retains members is what the 90-day POC proves.

€2,000 buys the proof; not a subscription for something unproven.

Dina Bosma-Buczynska · AI Consultancy Bootcamp · March 2026